

Windy conditions

Arrival 10:46 LT

Maintenance sheet version 10 for accumulation area (single mast, two levels)

|                |                                                                  |
|----------------|------------------------------------------------------------------|
| Station name   | NAU                                                              |
| People present | HETJA, APA, BAV, AEU, Pilots Johann + Greta                      |
| Purpose        | Maintenance / installation / take-down / quick visit / emergency |
| Date           | 10/05/2025                                                       |

Notes: Old CIBES mast sawed down and removed, to be out of the way for Twin Otter landing next year  
 - Two Vaisalas installed + new loggerbox + new thermistor string in PVC pipe

| AWS to do list                        |                     | Check after maintenance |                                                                                                                                                                                             |                  |                                                                         |
|---------------------------------------|---------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------|
| Part                                  | Shipped? (Y/N)      | Replaced? (Y/N)         | Number shipped/arrival                                                                                                                                                                      | Number departure |                                                                         |
| Radiometer                            | Y                   | Y                       | SN:171267<br>Calib: 10/04/2024                                                                                                                                                              | 171267           | Clean? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N |
| Iridium modem IMEI number             | N                   |                         | (30023406)                                                                                                                                                                                  | 1165 160         |                                                                         |
| Inclinometer/compass                  | N                   | <del>Y</del> Y          | Attached to the radiometer? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N                                                                                                |                  |                                                                         |
| Satellite antenna                     | N                   | N                       |                                                                                                                                                                                             |                  |                                                                         |
| Wind sensor upper                     | Y                   | Y                       | Black box pointing toward the radiometer (i.e., along the boom)? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N                                                           |                  |                                                                         |
| Wind sensor lower                     | Y                   | Y                       | Black box pointing same direction as the radiometer? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N<br>Wind sensor is one brought back from CPZ                           |                  |                                                                         |
| Temperature / humidity assembly upper | Y (Lufft + Vaisala) | Y                       | Fan spins? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N<br>Sounds, OK? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N<br>Lufft beslag er non standard |                  |                                                                         |
| Temperature / humidity assembly lower | Y (Lufft) Vaisala   | Y Vaisala               | Fan spins? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N<br>Sounds, OK? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N                                 |                  |                                                                         |
| Sonic ranger upper                    | Y                   | Y                       | Old sensor with new membrane? Y / <input type="checkbox"/> N                                                                                                                                |                  |                                                                         |
| Sonic ranger lower                    | Y                   | Y                       | Old sensor with new membrane? Y / <input type="checkbox"/> N                                                                                                                                |                  |                                                                         |
| Thermistor string                     | Y                   | Y                       | see comment below                                                                                                                                                                           |                  |                                                                         |
| Solar panel                           | Y                   | Y                       | Clean? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N<br>NB: is not effective when partially covered                                                                      |                  |                                                                         |
| GPS antenna                           | N                   | N                       | At top of logger enclosure? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N                                                                                                |                  |                                                                         |
| Logger box ID                         | Y<br>M4-08          | Y                       | CR1000 or <u>CR1000X?</u><br>ID number: <u>M4-008</u>                                                                                                                                       |                  |                                                                         |
| Logger enclosure                      | N                   | N                       | Replaced (including everything inside)? Y / N<br>New desiccant bags? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N                                                       |                  |                                                                         |
| Battery box                           | Y                   | Y                       | New box? <input checked="" type="checkbox"/> Y <input type="checkbox"/> N<br>New desiccant bags? Y / <input type="checkbox"/> N<br>No desiccant bags                                        |                  |                                                                         |
| Mast extension inserted?              | Y                   | Y                       | How many inserted? How long? 1 standard 1.5m<br>Notes:                                                                                                                                      |                  |                                                                         |
| Mast                                  | N                   | N                       | New mast? Y / <input type="checkbox"/> N<br>New sensor boom? Y / <input type="checkbox"/> N                                                                                                 |                  |                                                                         |

Comment: Thermistor string installation difficult. Blowing snow had clogged the individual pieces. String would not pass through PVC pipe so 12m pipe was taken out and disassembled, each piece cleaned for snow, string pushed through one piece at the time. Then the pipe was put back in the hole in 4m pieces with string pre-inserted.  
 Realizing we could not determine depth of top sensor (should be at surface) we tried to pull the 12m PVC pipe a little up to reach top connection at 10m. Unfortunately the pipe came apart down in the hole as we did this, but it

Note: GNNs mounted on mast at 0 cm (zero) from bottom of mount to mast connection and left on for >>30 min with mastwork and snowpit at some distance away

Arr 10:46 LT

WINDY COND IT19

|              |             |                    |
|--------------|-------------|--------------------|
| Station name | NAU         | UTC: 11:46 - 18:46 |
| Date         | 10 MAY 2025 | 10:46 - 17:46 LT   |

#### Metadata maintenance

HEDA, APA, BAV, ABU, Joham, Greta

|                                                                                    | Arrival                                                                        | Departure                                                          |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Time difference between logger clock and UTC ( $\pm$ mm:ss)                        | +00:03                                                                         | 00:00                                                              |
| Download data to PC and/or change SD card (wait for green light)                   | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N             | <input type="checkbox"/> Y / <input checked="" type="checkbox"/> N |
| Name of logger program                                                             | GCNET2_A                                                                       | GCNET3_2                                                           |
| Screen dump of values in fast scan mode                                            | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N             | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N |
| Photos of mast, sensors, logger box wiring etc. (more is better)                   | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N             | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N |
| Upper boom direction with radiometer relative to north                             | $^{\circ}$ true/magn.                                                          | $^{\circ}$ true/magn.                                              |
| Lower boom perpendicular to upper boom.<br>If not, measure.                        | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N             | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N |
| Box on upper wind sensor exactly same direction as radiometer?<br>If not, measure. | points East <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N |
| Box on lower wind sensor exactly same direction as radiometer?<br>If not, measure. | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N             | <input checked="" type="checkbox"/> Y / <input type="checkbox"/> N |
| Upper temperature / humidity height (bottom of casing - surface)                   | $\sim 269$ cm                                                                  | obs. not done cm                                                   |
| Lower temperature / humidity height (bottom of casing - surface)                   | 124 cm                                                                         | 270 cm                                                             |
| Upper sonic ranger height (bottom of casing - surface)                             | 229 cm                                                                         | 372 cm                                                             |
| Lower sonic ranger height (bottom of casing - surface)                             | 103 cm                                                                         | 246 cm                                                             |
| Length of thermistor string on surface from surface marking                        | N/A cm                                                                         | N/A cm                                                             |
| Apparent damage to sensors? If yes, which?                                         | <input type="checkbox"/> Y / <input checked="" type="checkbox"/> N             | <input type="checkbox"/> Y / <input checked="" type="checkbox"/> N |

#### Notes:

- snow in lower Luftt prep
- Lower wind sensor broke during installation, so an old one from CP1 was used instead.
  - propeller of lower wind sensor broke during raising of mast a new spare one was installed
  - Paper inside logger box which shows connector number and function blew away

#### Must-do at next visit:

- Normal "beslag" for Luftt sensor
- Normal "beslag" for lower wind sensor
- It may not be possible to retrieve/pull up the thermistor string (see comment on prev. page), so a full spare thermistor + PK pipe + cable should be considered
- New paper for inside the logger box, describing the connector function